

Feb22

Gao

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```
library(datasets)
library(tidymodels)
```

```
## -- Attaching packages ----- tidymodels 1.1.1 --
```

```
## v broom      1.0.5      v recipes      1.0.9
## v dials      1.2.0      v rsample     1.2.0
## v dplyr      1.1.4      v tibble     3.2.1
## v ggplot2    3.4.4      v tidyr      1.3.1
## v infer      1.0.6      v tune       1.1.2
## v modeldata  1.3.0      v workflows  1.1.3
## v parsnip    1.1.1      v workflowsets 1.0.1
## v purrr      1.0.2      v yardstick  1.3.0
```

```
## -- Conflicts ----- tidymodels_conflicts() --
```

```
## x purrr::discard() masks scales::discard()
## x dplyr::filter()   masks stats::filter()
## x dplyr::lag()      masks stats::lag()
## x recipes::step()   masks stats::step()
## * Use tidymodels_prefer() to resolve common conflicts.
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v forcats  1.0.0      v readr      2.1.5
## v lubridate 1.9.3      v stringr    1.5.1
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x readr::col_factor() masks scales::col_factor()
## x purrr::discard()    masks scales::discard()
## x dplyr::filter()     masks stats::filter()
## x stringr::fixed()    masks recipes::fixed()
## x dplyr::lag()         masks stats::lag()
## x readr::spec()       masks yardstick::spec()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(rpart.plot)
```

```
## Loading required package: rpart
##
## Attaching package: 'rpart'
##
## The following object is masked from 'package:dials':
##
##     prune
```

```
library(caret)
```

```
## Loading required package: lattice
##
## Attaching package: 'caret'
##
## The following objects are masked from 'package:yardstick':
##
##     precision, recall, sensitivity, specificity
##
## The following object is masked from 'package:purrr':
##
##     lift
```

```
library(ggforce)
```

```
irises <- as_tibble(iris)
```

```
irises
```

```
## # A tibble: 150 x 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##         <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

```
irises$Species %>% unique()
```

```
## [1] setosa    versicolor virginica
## Levels: setosa versicolor virginica
```

```
iris_split <- initial_split(irises, prop = 0.75)
```

```
train <- training(iris_split)
```

```
test <- testing(iris_split)
```

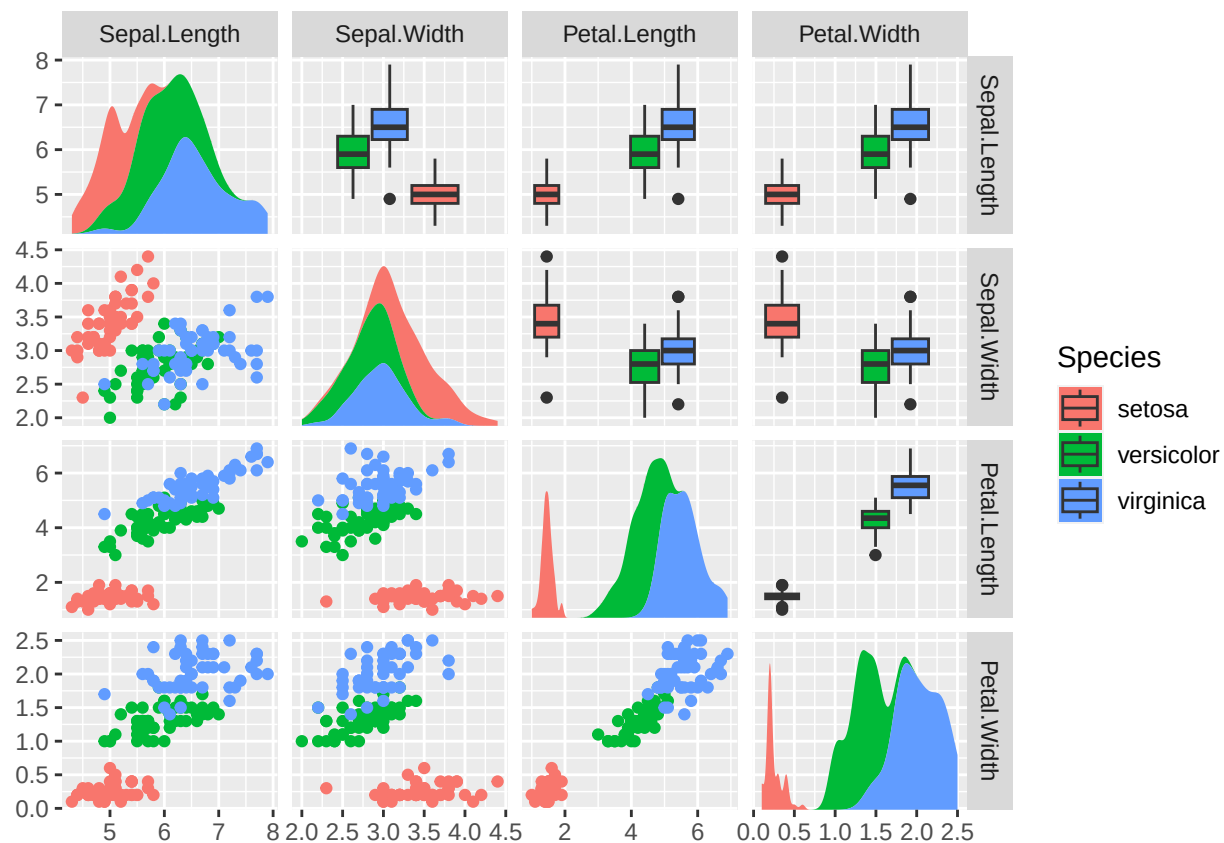
```
train
```

```
## # A tibble: 112 x 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1         5.7         3         4.2         1.2 versicolor
## 2         6         3         4.8         1.8 virginica
## 3         5.8         2.8         5.1         2.4 virginica
## 4         7.1         3         5.9         2.1 virginica
## 5         5.4         3         4.5         1.5 versicolor
## 6         5         2         3.5         1 versicolor
## 7         4.7         3.2         1.3         0.2 setosa
## 8         6.3         2.5         5         1.9 virginica
## 9         5         3.5         1.6         0.6 setosa
## 10        5.2         3.5         1.5         0.2 setosa
## # i 102 more rows
```

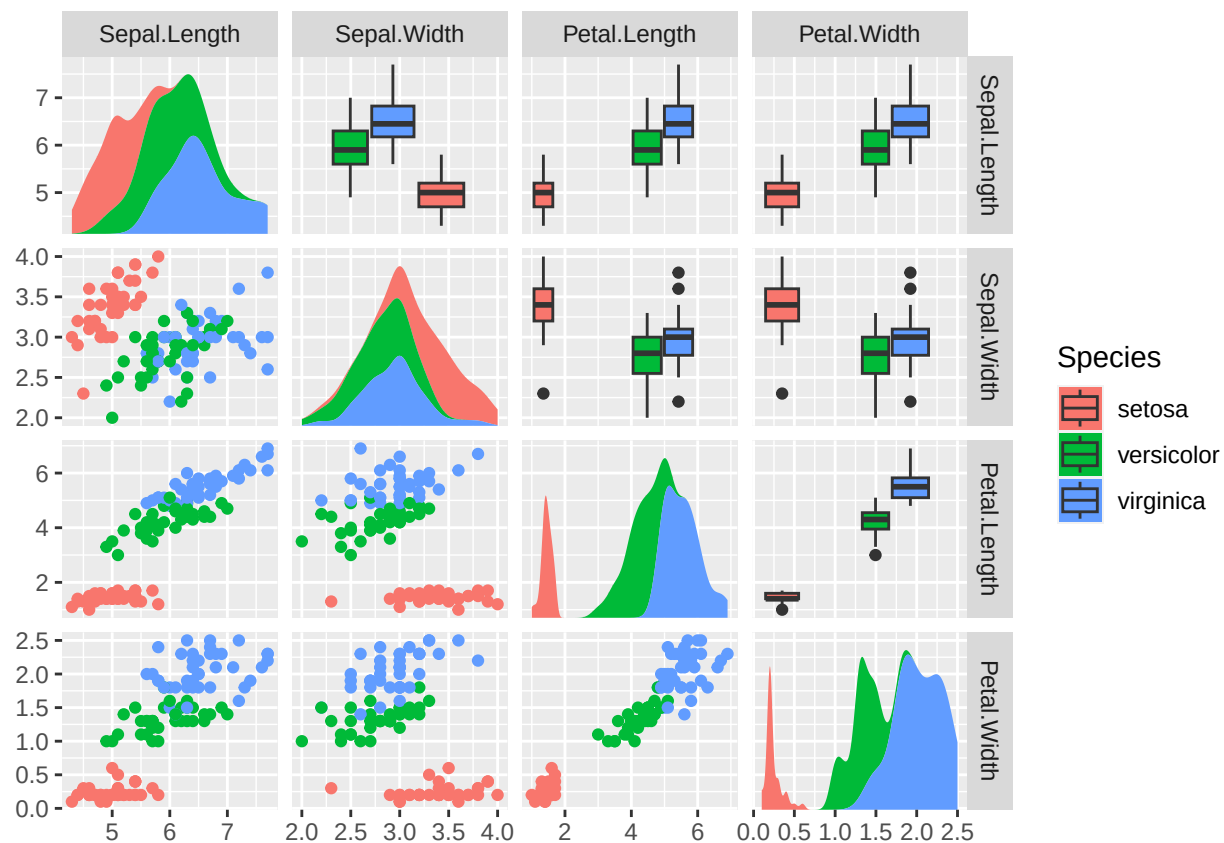
```
test
```

```
## # A tibble: 38 x 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1         4.9         3.1         1.5         0.1 setosa
## 2         5.7         4.4         1.5         0.4 setosa
## 3         5.1         3.7         1.5         0.4 setosa
## 4         4.8         3.4         1.9         0.2 setosa
## 5         5         3.4         1.6         0.4 setosa
## 6         5.2         4.1         1.5         0.1 setosa
## 7         5.5         4.2         1.4         0.2 setosa
## 8         4.9         3.1         1.5         0.2 setosa
## 9         5         3.2         1.2         0.2 setosa
## 10        4.4         3         1.3         0.2 setosa
## # i 28 more rows
```

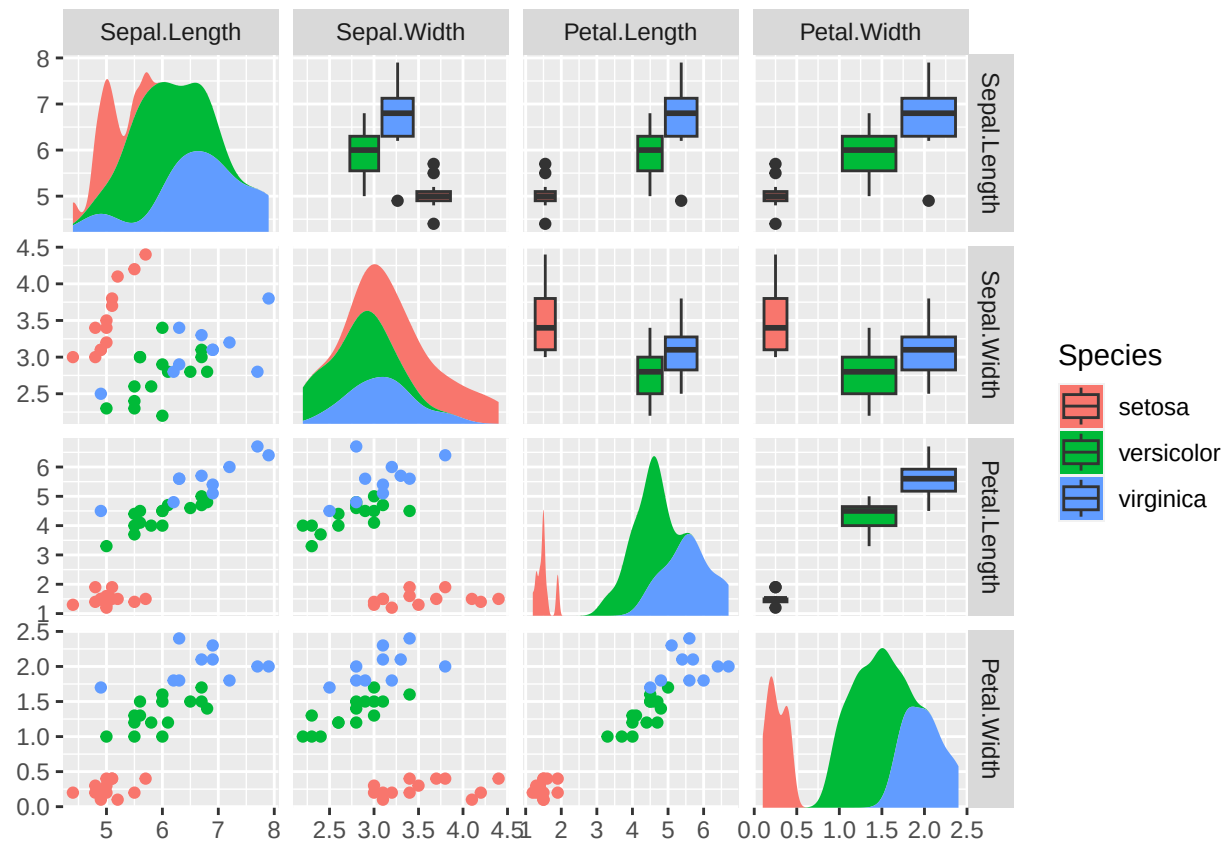
```
ggplot(data = irises, aes(x = .panel_x, y = .panel_y)) +
  geom_point(aes(color = Species)) +
  geom_autodensity(aes(fill = Species)) +
  geom_boxplot(aes(fill = Species)) +
  facet_matrix(vars(Sepal.Length:Petal.Width),
    layer.lower = 1,
    layer.diag = 2,
    layer.upper = 3)
```



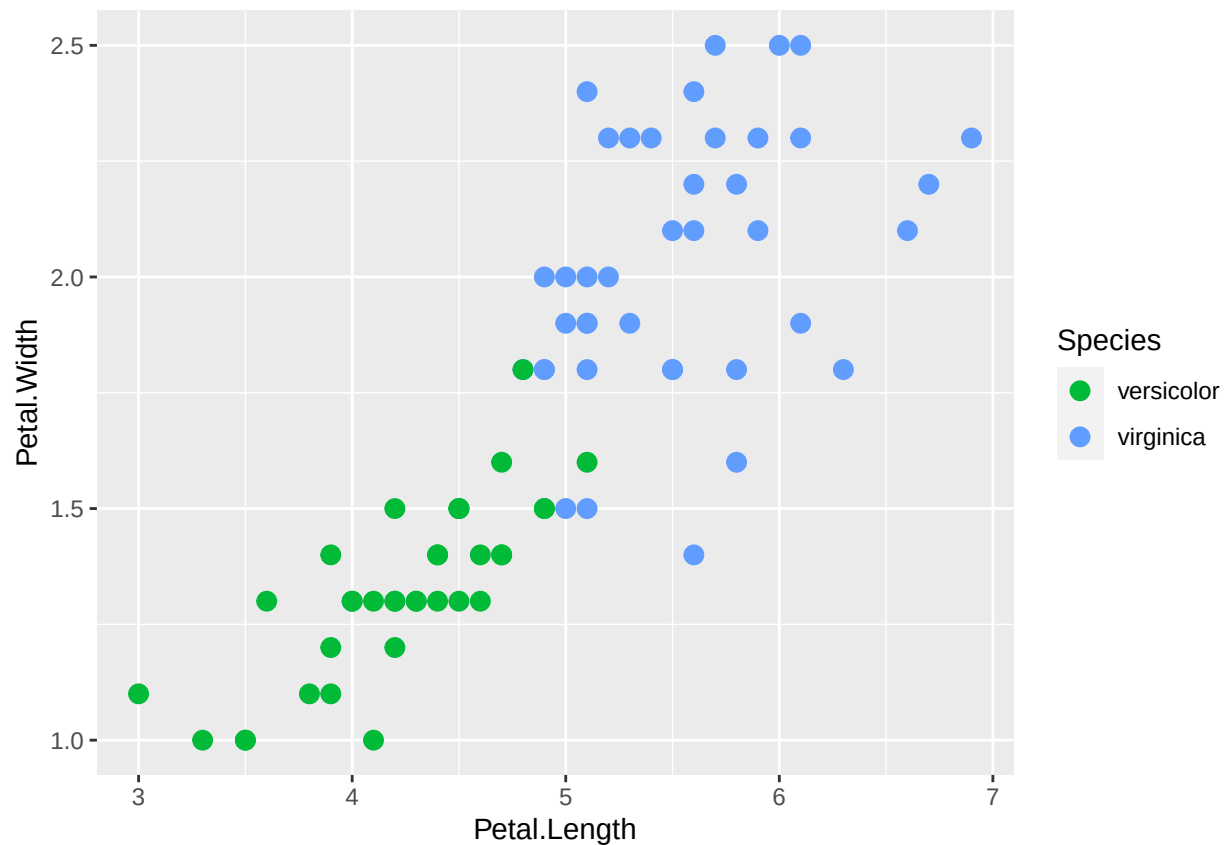
```
ggplot(data = train, aes(x = .panel_x, y = .panel_y)) +
  geom_point(aes(color = Species)) +
  geom_autodensity(aes(fill = Species)) +
  geom_boxplot(aes(fill = Species)) +
  facet_matrix(vars(Sepal.Length:Petal.Width),
    layer.lower = 1,
    layer.diag = 2,
    layer.upper = 3)
```



```
ggplot(data = test, aes(x = .panel_x, y = .panel_y)) +
  geom_point(aes(color = Species)) +
  geom_autodensity(aes(fill = Species)) +
  geom_boxplot(aes(fill = Species)) +
  facet_matrix(vars(Sepal.Length:Petal.Width),
    layer.lower = 1,
    layer.diag = 2,
    layer.upper = 3)
```

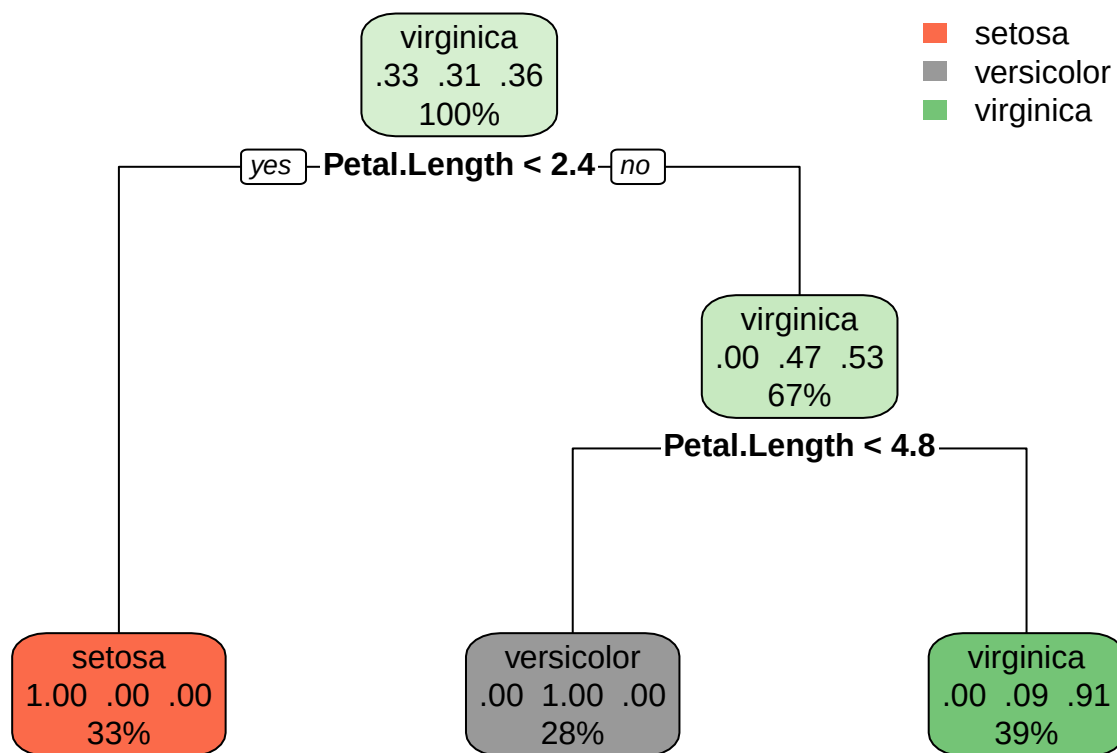


```
ggplot(filter(train, Species != "setosa")) + geom_point(aes(x = Petal.Length, y = Petal.Width, color = Species))
```



```
dtree_mod <- decision_tree(mode = "classification", min_n = 4, tree_depth = 3)
dtree_mod_fit <- fit(object = dtree_mod, formula = Species ~ ., data = train)
dtree_mod_fit %>% extract_fit_engine() %>% rpart.plot()
```

```
## Warning: Cannot retrieve the data used to build the model (so cannot determine roundint and is.binary)
## To silence this warning:
##   Call rpart.plot with roundint=FALSE,
##   or rebuild the rpart model with model=TRUE.
```



```
predictions <- predict(dtrees_mod_fit, new_data = test)
```

```
confusionMatrix(data = predictions$.pred_class, reference = test$Species)
```

```
## Confusion Matrix and Statistics
```

```
##
```

```
##           Reference
```

```
## Prediction  setosa versicolor virginica
```

```
##   setosa      13         0         0
```

```
##   versicolor   0        13         1
```

```
##   virginica    0         2         9
```

```
##
```

```
## Overall Statistics
```

```
##
```

```
##           Accuracy : 0.9211
```

```
##           95% CI : (0.7862, 0.9834)
```

```
##   No Information Rate : 0.3947
```

```
##   P-Value [Acc > NIR] : 1.467e-11
```

```
##
```

```
##           Kappa : 0.8806
```

```
##
```

```
##   McNemar's Test P-Value : NA
```

```
##
```

```
## Statistics by Class:
```

```
##
```

##	Class: setosa	Class: versicolor	Class: virginica
## Sensitivity	1.0000	0.8667	0.9000
## Specificity	1.0000	0.9565	0.9286
## Pos Pred Value	1.0000	0.9286	0.8182
## Neg Pred Value	1.0000	0.9167	0.9630
## Prevalence	0.3421	0.3947	0.2632
## Detection Rate	0.3421	0.3421	0.2368
## Detection Prevalence	0.3421	0.3684	0.2895
## Balanced Accuracy	1.0000	0.9116	0.9143